

Truly Innovative **TECH**

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Tech*

'5-minute charge' EV batteries

StoreDot's extreme fast charging (XFC) batteries for two-wheel electric vehicles (EVs) charge fully within five minutes and can thus eliminate range and charging issues thanks to their small form-factor battery cell. This is achieved by

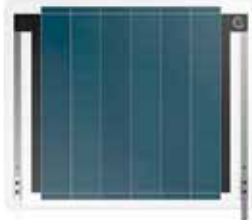


replacing graphite in the Li-ion battery cell's anode with metalloid nano-particles—a key breakthrough in overcoming major issues in safety, battery cycle life, and swelling. Besides EVs, the ultra-fast charging batteries can be used for commercial drones and consumer electronics. The XFC batteries comply with UN 38.3 standard, which ensures safety during shipping.

StoreDot
www.store-dot.com

Flexible organic solar cell

Epishine's 'light cell' is a thin and flexible organic solar cell that can be integrated in sensors, consumer electronics, and other low power devices to reduce or eliminate the use of batteries.



The new light energy harvesting cells are non-toxic, based on organic electronics, and encapsulated in recyclable plastics. The light cells are optimised for ambient light indoors. All electronic devices powered by small batteries that last for a year or more can potentially be

powered by harvesting ordinary indoor light with this innovation. Their manufacturing process is based on different printing techniques, roll-to-roll.

Epishine
www.epishine.com

Vehicle passengers' 4D safety solution

Vayyar's multifunctional single-sensor platform addresses the challenging current and upcoming safety requirements, offering vehicles up to ten in-cabin points while dramatically reducing complexity and costs. The lifesaving



single-sensor solution offers child presence detection plus enhanced seat-belt reminders. Its high-resolution 4D data enables all OEMs and Tier 1s to independently develop advanced safety and comfort applications over the platform, creating a safe in-cabin ecosystem. One high-performance 4D imaging radar-on-chip covers the entire cabin, reducing hardware, wiring and overall costs, while replacing seven in-car sensors.

Vayyar
www.vayyar.com

Integrated solution for audio video bridging

Microchip's LAN9360 audio endpoint controller interconnects vehicles' infotainment devices including speakers, amplifiers, microphones, navigation systems, radio tuners, and smart headrests with Ethernet AVB. It bridges audio between Ethernet AVB and inter-IC sound, time division multiplexing, and pulse density modulation local audio interfaces. LAN9360 completely sup-

ports audio transmission over Ethernet AVB, including generalised precision time protocol, timestamping, transport protocols, and content protection with high-bandwidth digital content protection. It also supports secure boot and secure



remote updates over Ethernet. It requires no software integration, enabling designers to configure the device simply and quickly to manufacturers' unique audio and networking requirements.

MicrochipTechnology
www.microchip.com

Compact odour imaging sensor

Aroma Bit has developed the world's first compact odour imaging sensor that digitises odours with visualisation patterns, similar to an animal's sense of smell. It differs from conventional gas



sensors, which can only detect specific substances. The device has a wide range of applications, including the food, agriculture, daily commodities, industrial machinery, mobility, healthcare, robotics, mobile information service, home electronics, and IoT industries. The silicon CMOS sensor module development kit 5C-SSM-H1 launched in January features 80 elements arranged on a 6mm × 3mm silicon chip (5 types of odour-sensitive membrane × 16 elements per membrane).

Aroma Bit
www.aromabit.com